Q.29 Explain Carnot cycle with neat sketch.

Q.30 Define thermodynamic temperature scale.

Q.31 Explain heat pump and its working.

Q.32 Explain vapour compression refrigeration cycle.

Q.33 List properties of a good refrigerant.

Q.34 Explain Dalton's law of partial pressure.

Q.35 Explain Raoult's law.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain first law of thermodynamics for open system with neat diagram and example.

Q.37 Discuss and explain second law os thermodynamics with Kelvin-Plank and Clausius statements.

Q.38 Explain vapour absorption refrigeration system with neat and clean diagram.

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Roll No. ....

4th Sem / Chem, Chem Engg. (Spl. Paint Tech.)
Chem Engg. (Spl. Polymer Engg.)
Sub : Chemical Engg Thermodynamics / Engg Thermodynamics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Boundary of a system separates

a) System and surroundings

b) System and process

c) Surroundings only

d) Properties only

Q.2 Which of the following is a state function?

a) Work b) Heat

c) Internal energy d) Process

Q.3 Which of the following is an intensive property?

a) Mass b) Temperature

c) Volume d) Internal energy

Q.4 The unit of entropy is

a) J b) J/kg

c) J/k d) J/s

Q.5 Zeroth law of thermodynamics is concerned with

a) Energy transfer b) Temperature

c) Entropy d) Work

Q.6 For an ideal gas, enthalpy depends on

a) Pressure b) Volume

c) Temperature d) Mass

Q.7 Efficiency of Carnot engine depends upon

a) Working substance

b) Engine size

c) Temperatures of reservoirs

d) Pressure

Q.8 Entropy change for reversible adiabatic process is

a) Positive b) Negative

c) Maximum d) Zero

Q.9 COP of refrigerator is always

a) Less than 1 b) Equal to 1

c) Greater than 1 d) Zero

Q.10 The SI unit of temperature is

a) Celsius b) Fahrenheit

c) Kelvin d) Centigrade

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define thermodynamic system.

Q.12 Give one example of extensive property.

Q.13 Define path function.

Q.14 Write ideal gas equation.

Q.15 Define internal energy.

Q.16 State first law of thermodynamics.

Q.17 Define heat engine.

Q.18 What is entropy?

Q.19 Write expression of COP of refrigerator.

Q.20 Nam one eco friendly refrigerant.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Differentiate between open and closed system.

Q.22 Explain homogeneous and heterogeneous systems with examples.

Q.23 Discuss about intensive and extensive properties.

Q.24 Explain Joule's experiment and its significance.

Q.25 Explain isothermal process for ideal gas with equation.

Q.26 Write first law of thermodynamics for closed system.

Q.27 Explain limitations of first law of thermodynamics.

Q.28 Explain entropy change during phase change.